Name		Team	

Every sentence in the article is true. If you find yourself writing "this is wrong", check it again — it is probably true. The question is not whether it is true. The question is what it makes you believe.

The three sentences to look at

1 *"Team Comet's game was three times more popular than Team Nebula's."*

2 *"We asked four Prep students. Three said it was the best day of the year. That means 75% of Prep students rated Games Day the best day of the year."*

3 *"Twice as many Preps asked to play again next year as asked last year."*

Part A · What it makes you think

SF A

Sentence 1. Look at the graph. What are the two real numbers? Is one of them three times the other?

CF B

Sentence 2. How many Preps came to Games Day altogether? Why can four of them not speak for all of them?

CF C

Sentence 3. Twice as many as what? What would you need to be told before that sentence meant anything?

Frames: "This makes you think _____." · "What is actually true is _____." · "To know that, you would need _____."

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Redraw it fairly

Same three teams, same three numbers, axis starting at zero. Sketch it here — the real one goes in the spreadsheet. Title it and label the axis.

Team	Comet	Orion	Nebula
Buddies who liked it	11	10	9

WHAT CHANGED WHEN THE AXIS STARTED AT ZERO

WHAT DID NOT CHANGE

Frames: "When the axis started at zero, the bars _____." · "The numbers _____, and Comet was still _____."

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Exit ticket

On your own. Five minutes.

SF 1

A graph’s vertical axis starts at 40. Two bars reach 44 and 48. What is the real difference, and how many times taller does the second bar look?

CF 2

A poster says "9 out of 10 students prefer our canteen". Write the one question you would ask before you believed it.

CU 3

"Every student who ate breakfast got above 80% on the test. Breakfast makes you better at maths." Both sentences could be true. Explain why the second one still does not follow.

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